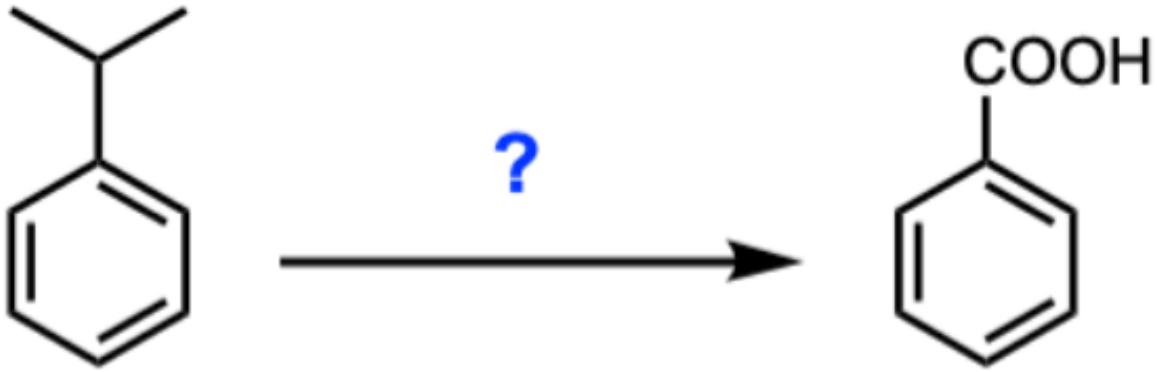
Q.1

Choose a suitable reagent to carry out this reaction :



(A) $NaBH_4$

(B) $H_2, Pd-C$

(C) $KMnO_4$

(D) H_2SO_3

Q.2

The oxidation of benzene by V_2O_5 in the presence of air produces:

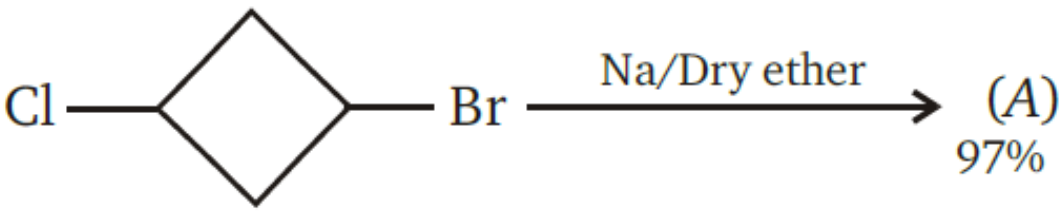
(A) benzoic acid

(B) benzaldehyde

(C) benzoic anhydride

(D) maleic anhydride

Q.3



Product (A) of above reaction is :

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(A)



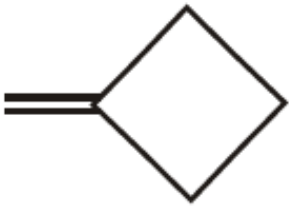
(B)



(C)

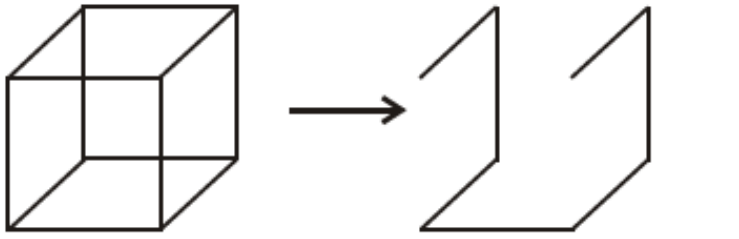


(D)



Q.4

How many bond cleavages are required to convert cubane into non-cyclic skeleton?



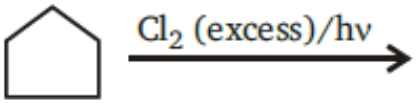
(A) 2

(B) 3

(C) 4

(D) 5

Q.5



How many dichloro products are formed in the above reaction (including stereoisomers)?

(A) 8

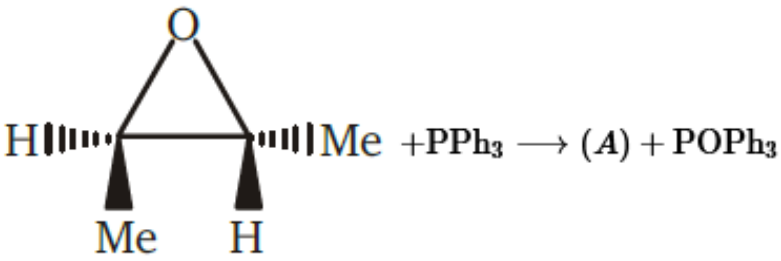
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(B) 6

(C) 7

(D) 9

Q.6



(major)

Product (A) is

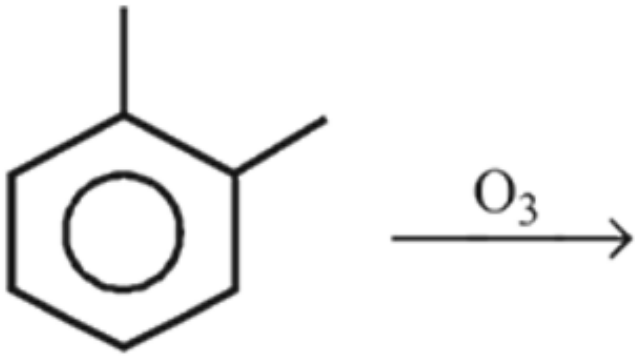
(A) trans-2-butane

(B) cis-2-butene

(C) 1-butene

(D) Iso-butene

Q.7



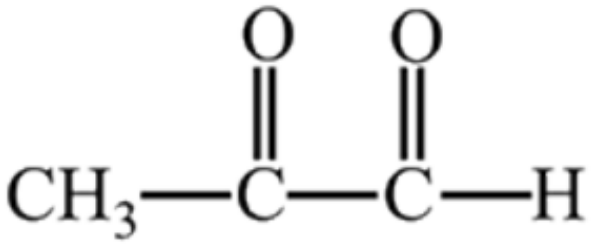
Which of the following products cannot be obtained in ozonolysis of o-xylene?

(A)

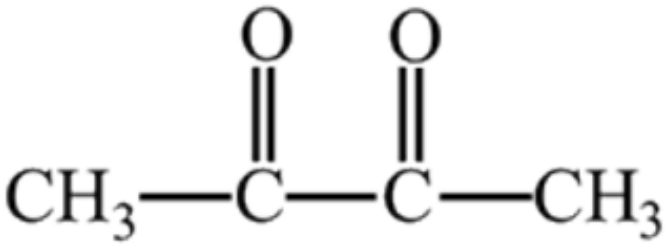


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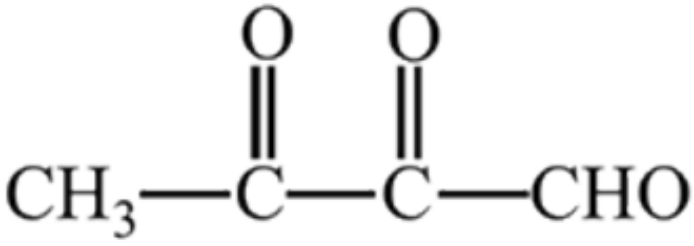
(B)



(C)



(D)



Q.8

Benzene on reaction with ozone forms

(A) 2 molecules of aldehyde and 1 molecule of ketone

(B) 2 molecules of ketone and 1 molecule of aldehyde

(C) triozonide

(D) hexaozonide

Q.9

$C_6H_5CH_3 \xrightarrow{CrO_2Cl_2} Z$

In the given sequence Z is:

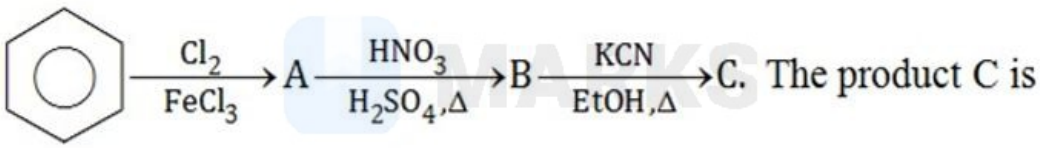
(A) benzaldehyde

(B) toluic acid

(C) phenyl acetic acid

(D) benzoic acid

Q.10



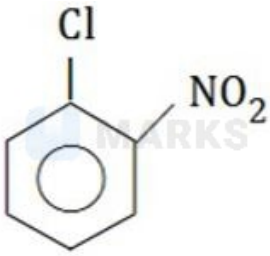
The product C is

@IITJEE_Advanced

(A)



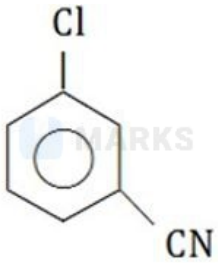
(B)



(C)

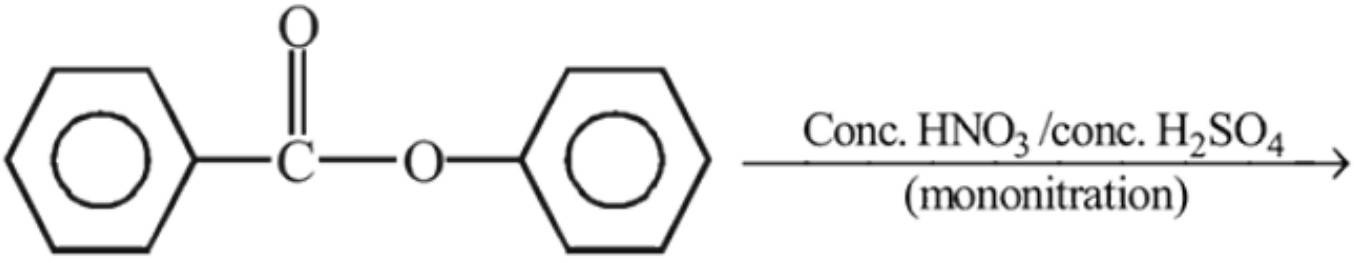


(D)

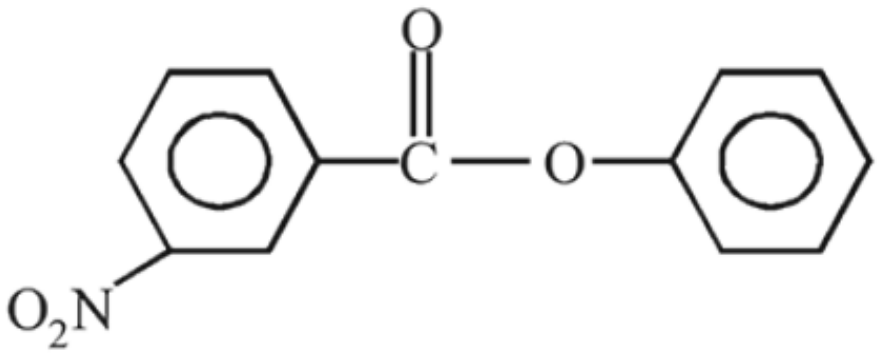


Q.11

The major product formed in the reaction is:

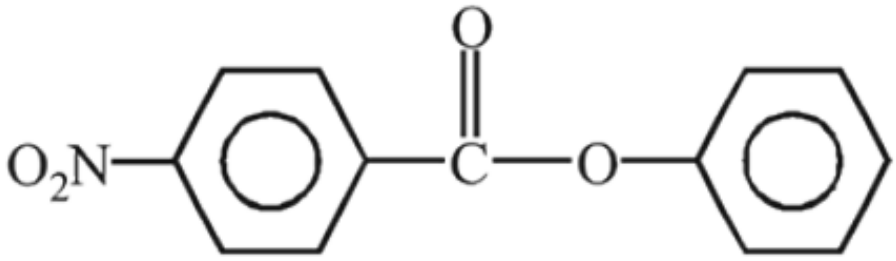


(A)

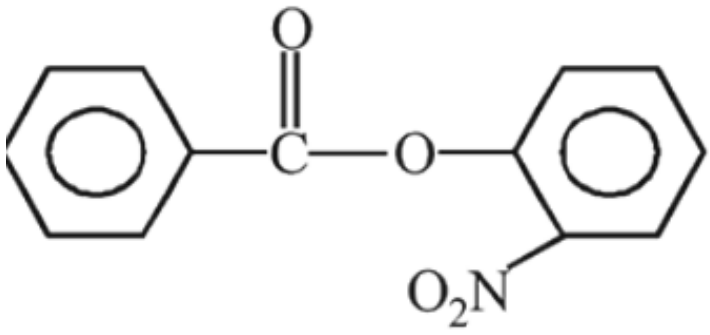


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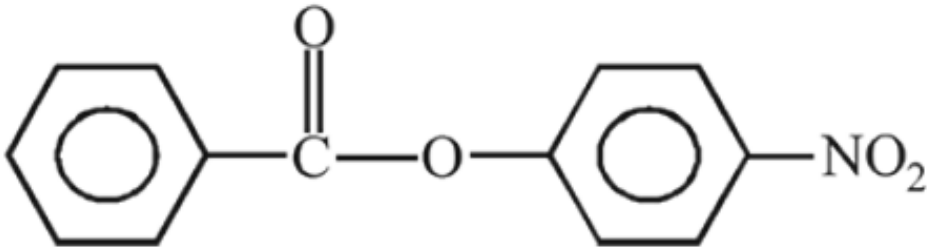
(B)



(C)

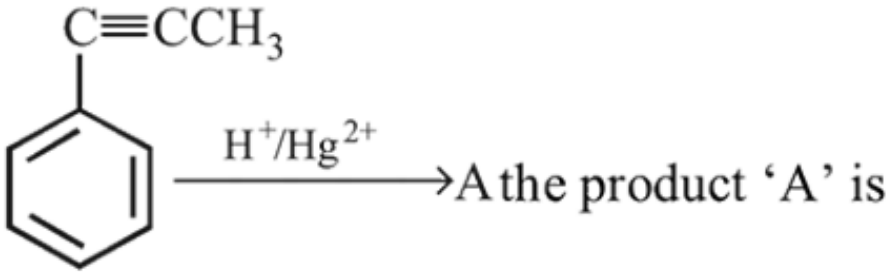


(D)

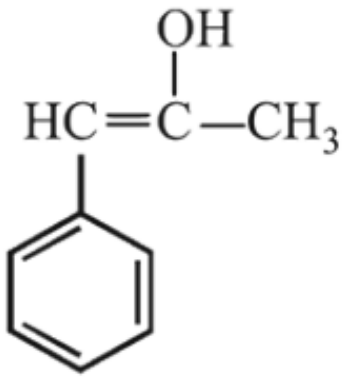


Q.12

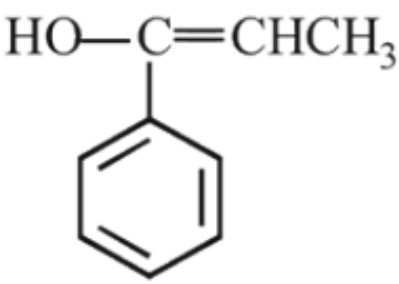
In the given reaction,



(A)

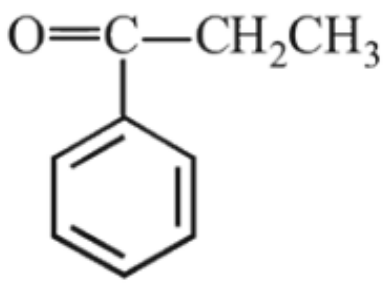


(B)

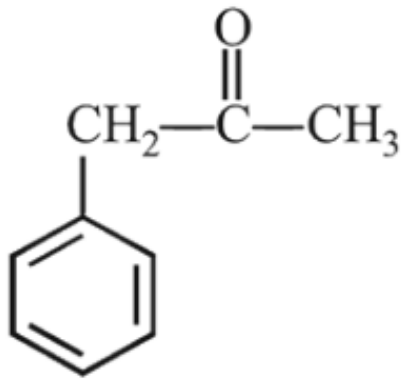


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(C)



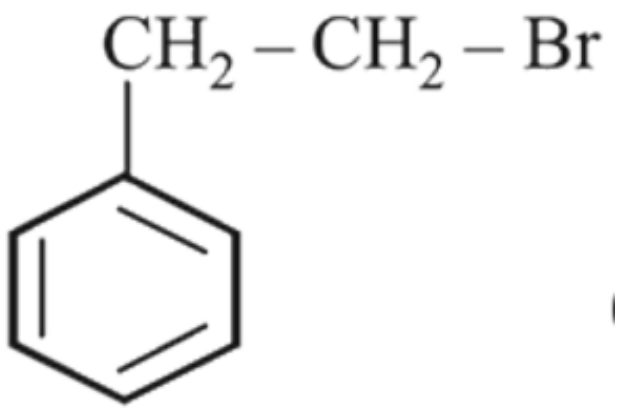
(D)



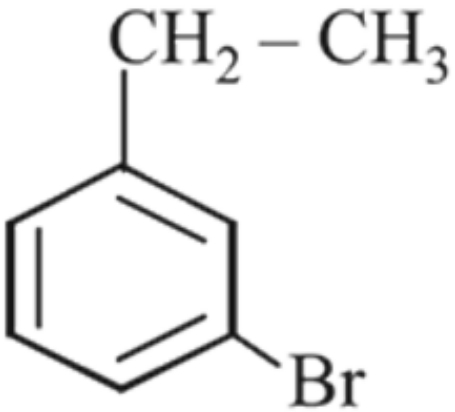
Q.13

The product of the reaction between ethyl benzene and N -bromosuccinamide is

(A)

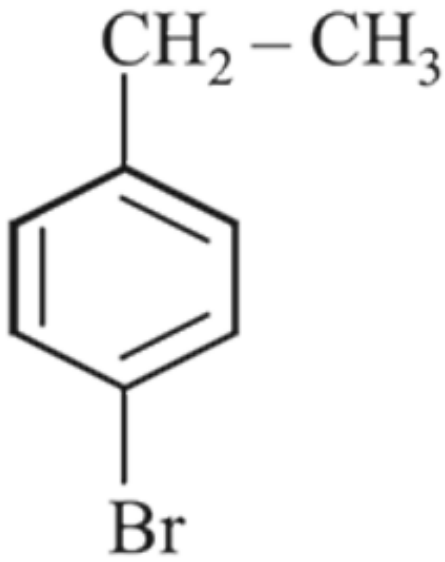


(B)

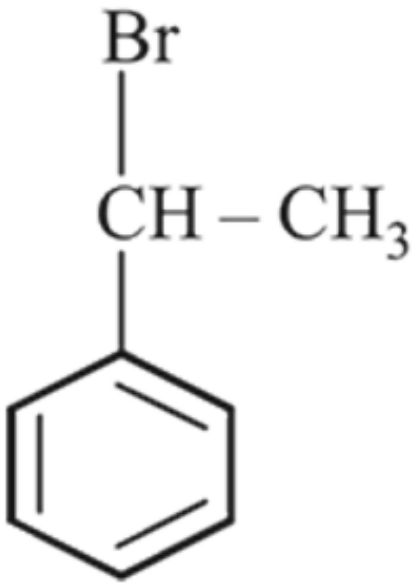


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(C)



(D)



Q.14

A group which deactivates the benzene ring towards electrophilic substitution but which directs the incoming group principally to the o- and p-positions is

(A) $-NH_2$

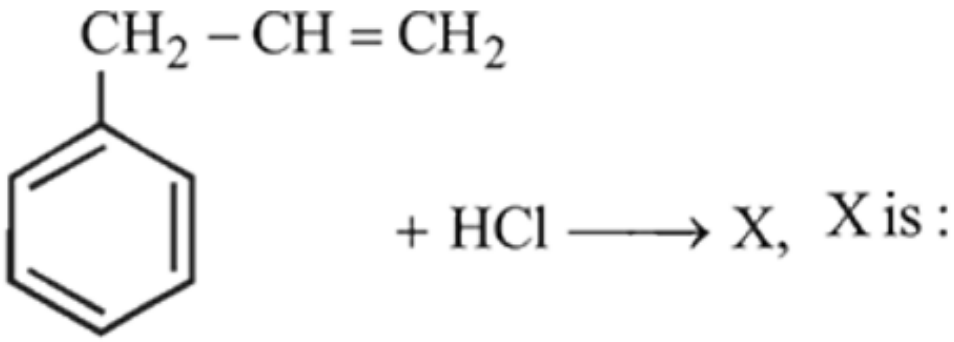
(B) $-Cl$

(C) $-NO_2$

(D) $-C_2H_5$

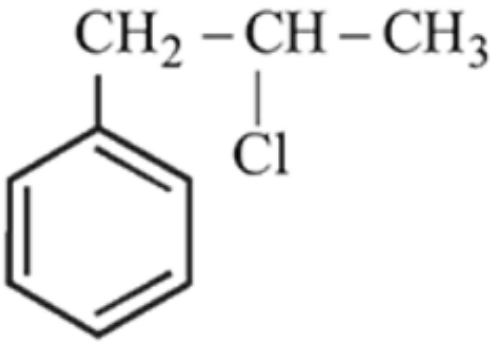
Q.15

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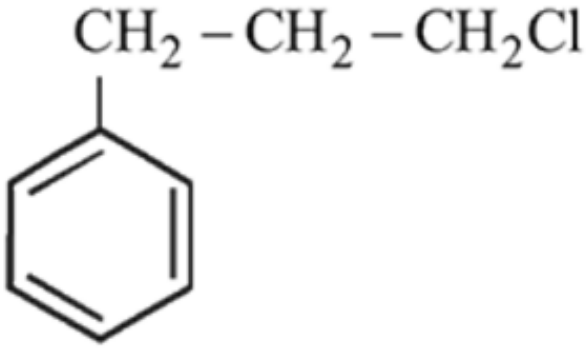


$+ HCl \longrightarrow X,$ X is :

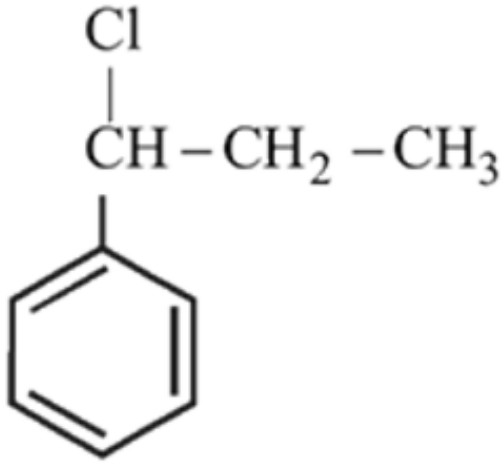
(A)



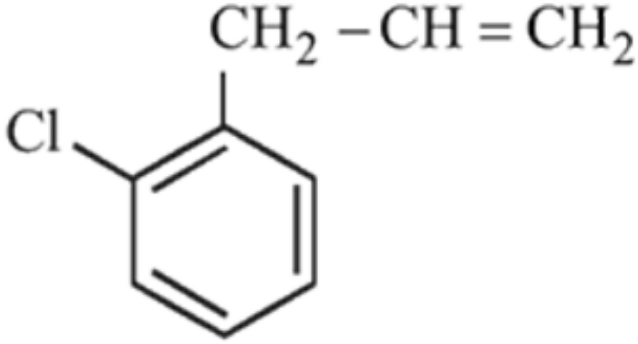
(B)



(C)



(D)



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Q.1 Answer:

$KMnO_4$

Solution:

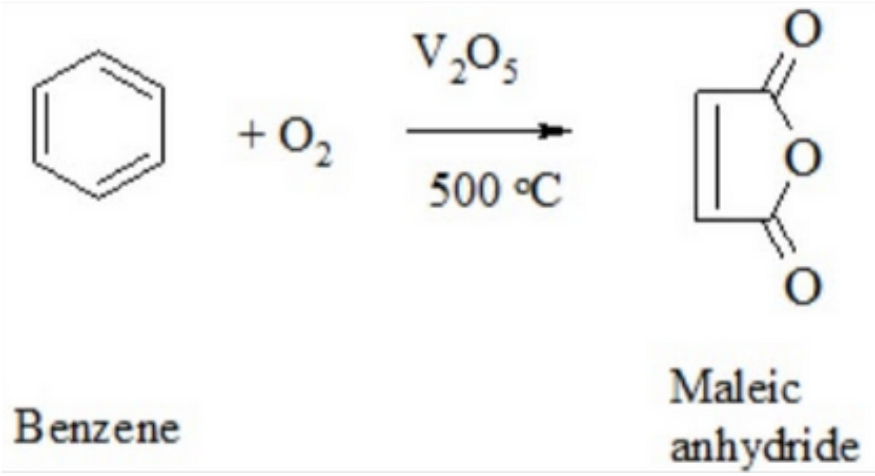
Treatment of an alkylbenzene with potassium permanganate results in oxidation to give the benzoic acid. The extra carbons are cleared and product benzoic acids is formed.

Q.2 Answer:

maleic anhydride

Solution:

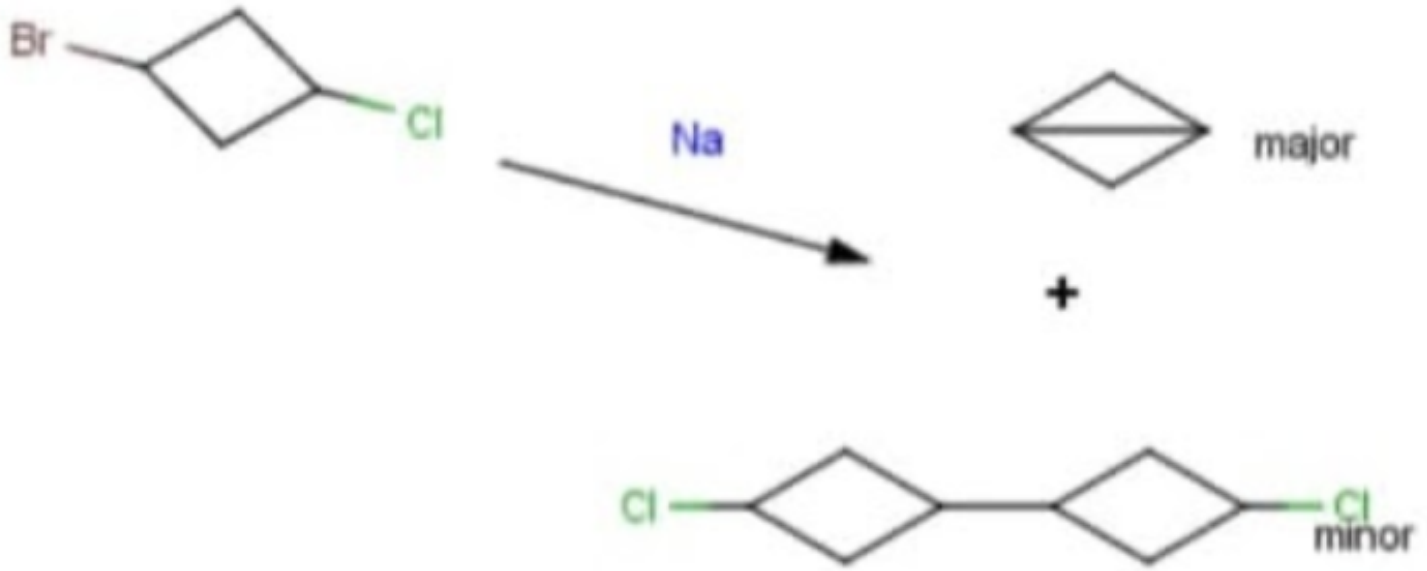
The oxidation of benzene by V_2O_5 in the presence of air produces maleic anhydride. The oxidation is carried out at $500^\circ C$



Q.3 Answer:



Solution:

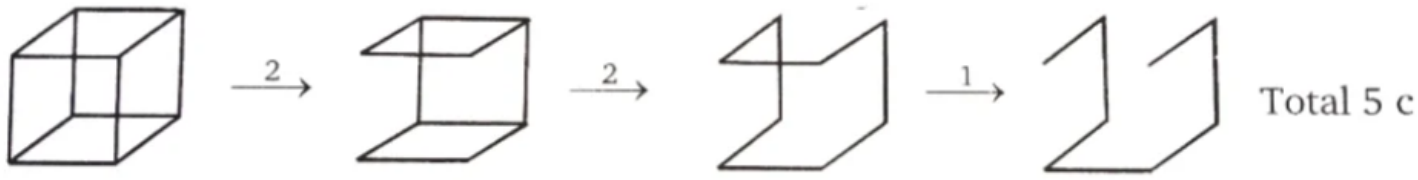


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Q.4 Answer:

5

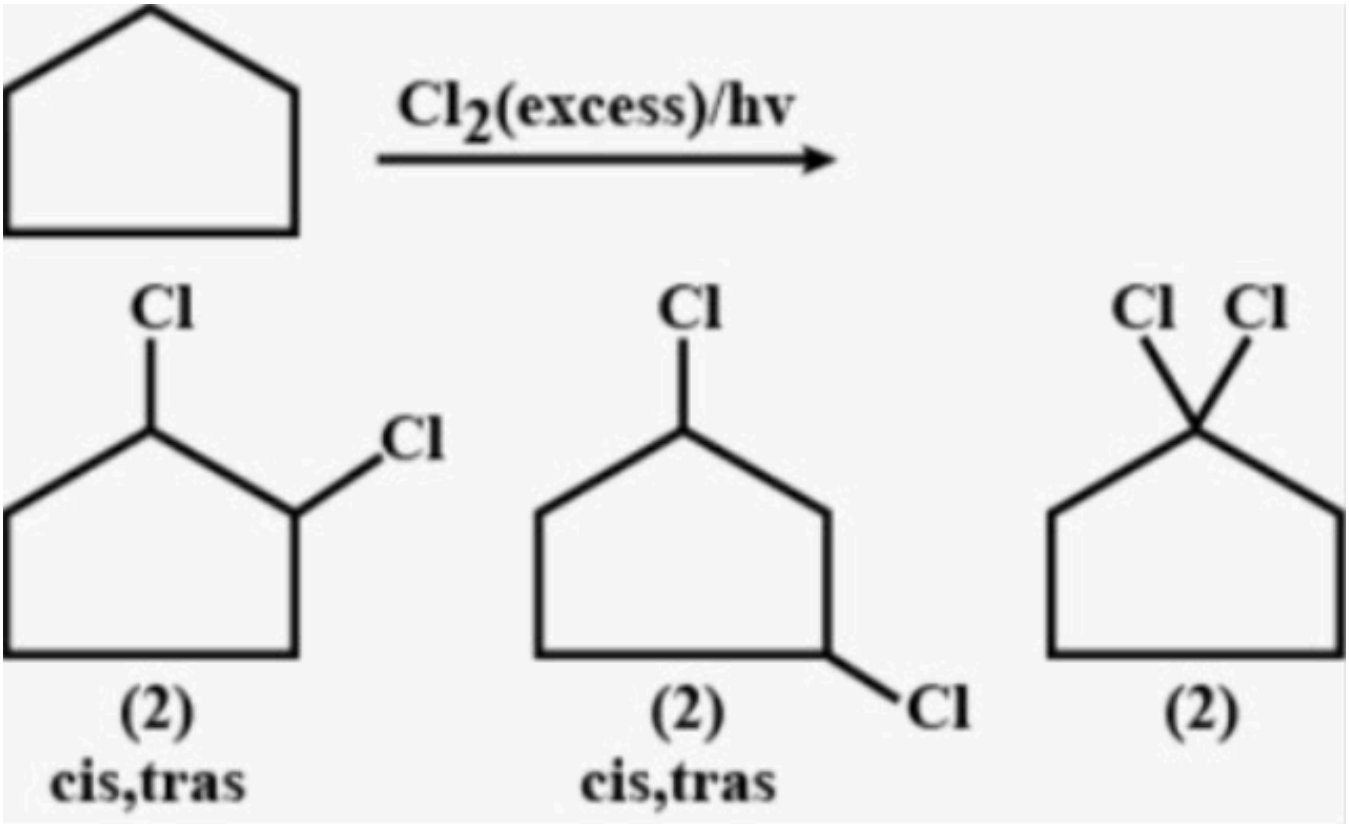
Solution:



Q.5 Answer:

8

Solution:



meso compound
so total product = 3

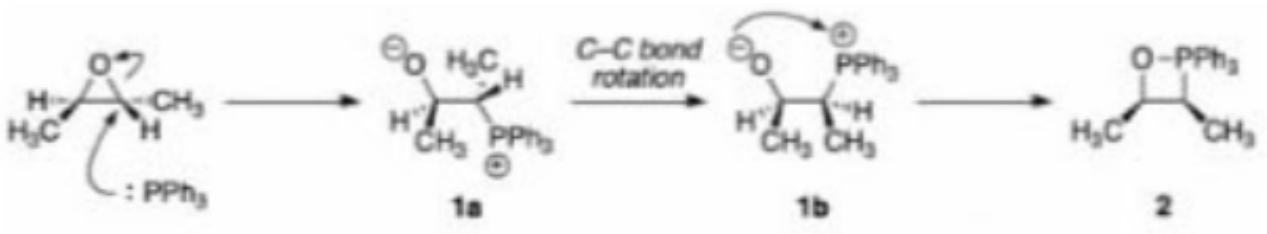
total number of
stereoisomers = 4

no stereocentre
present

Q.6 Answer:

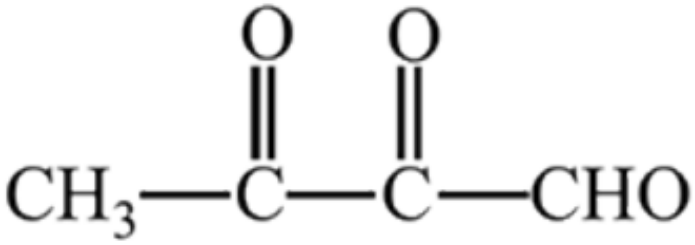
cis-2-butene

Solution:

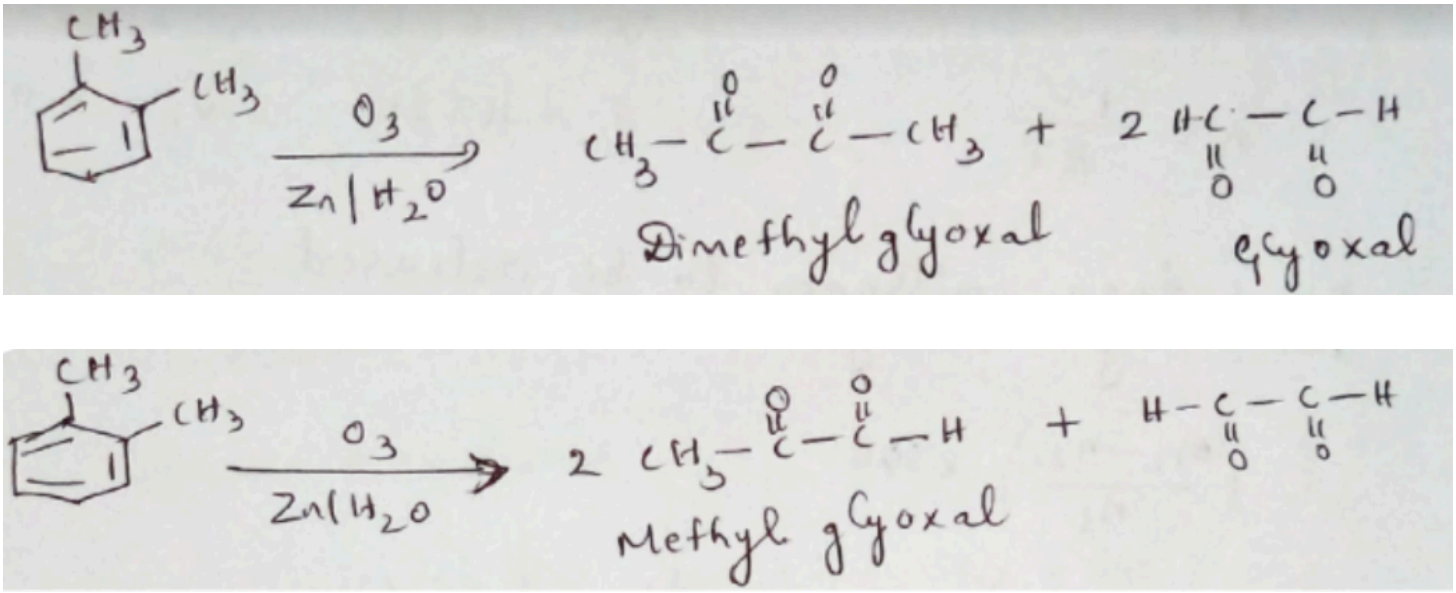


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Q.7 Answer:



Solution:

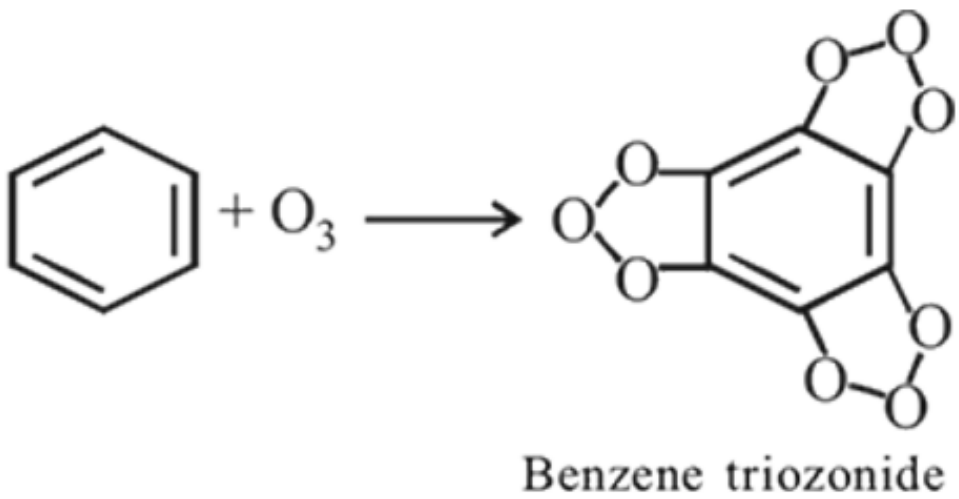


O-xylene will give a mixture of glyoxal , methyl glyoxal \& dimethyl glyoxal on ozonolysis. So, the product given in option (4) will not be obtained.

Q.8 Answer:

triozonide

Solution:



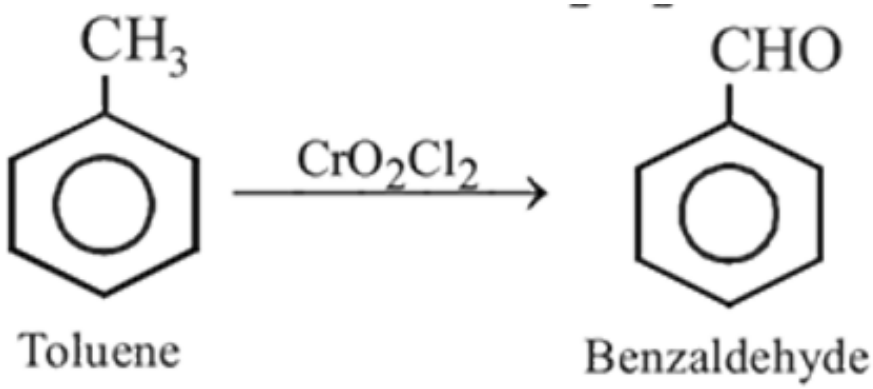
Q.9 Answer:

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benzaldehyde

Solution:

Toluene on oxidation with mild oxidising agent like chromyl chloride (CrO_2Cl_2) gives benzaldehyde.

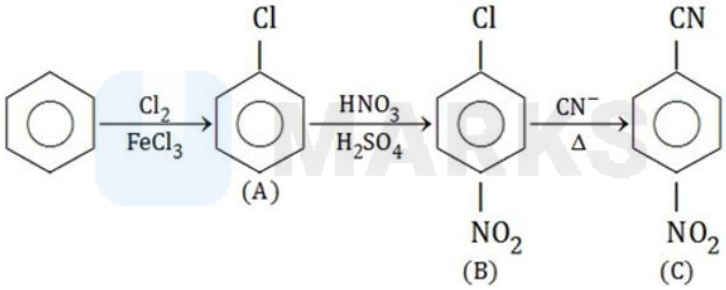


Q.10 Answer:

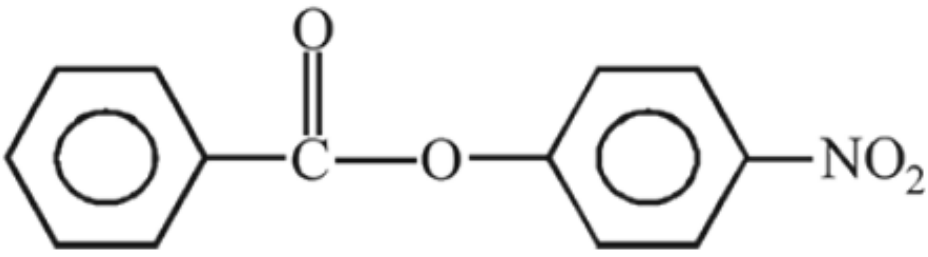


Solution:

The reaction sequence can be given as follows < br /> < br />

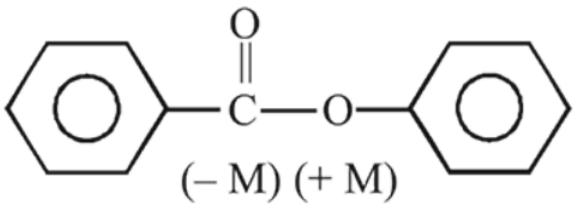


Q.11 Answer:



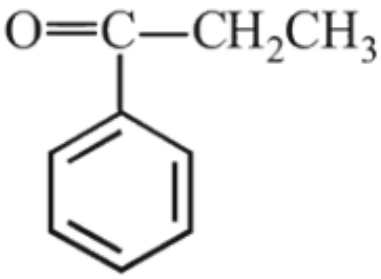
Solution:

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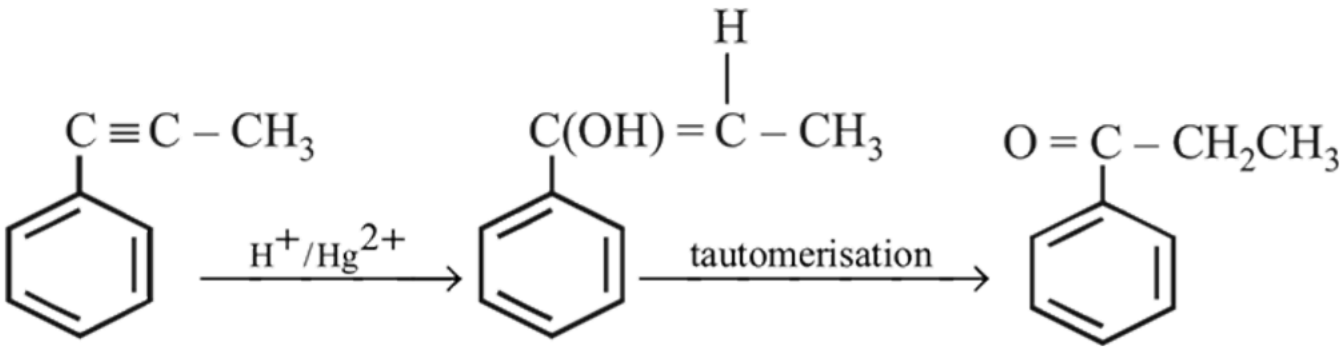


⟵ because of steric factor para is major.

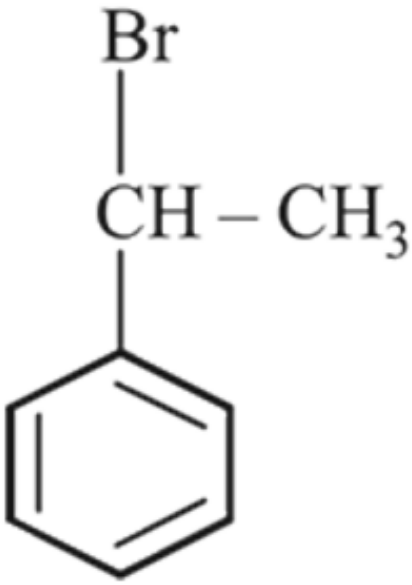
Q.12 Answer:



Solution:

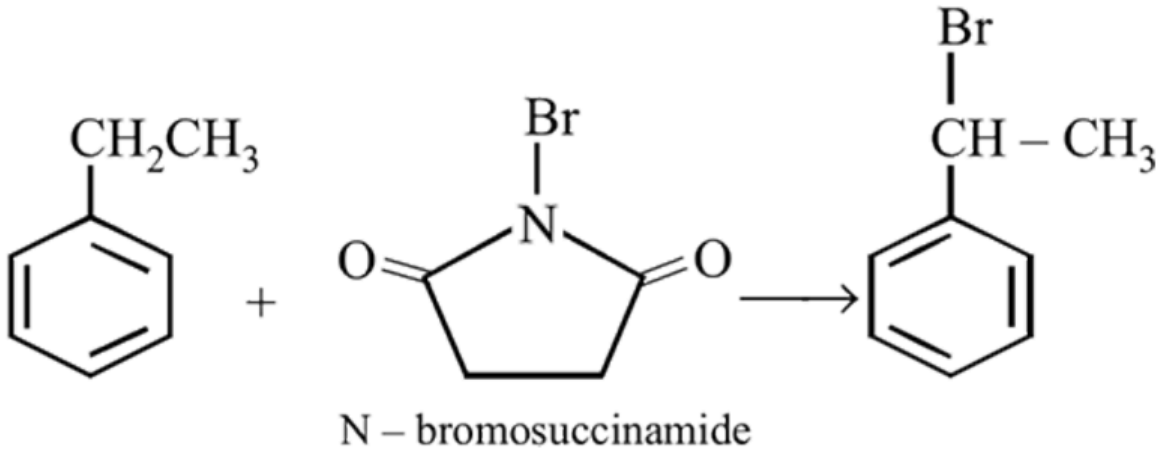


Q.13 Answer:



Solution:

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N – bromosuccinamide

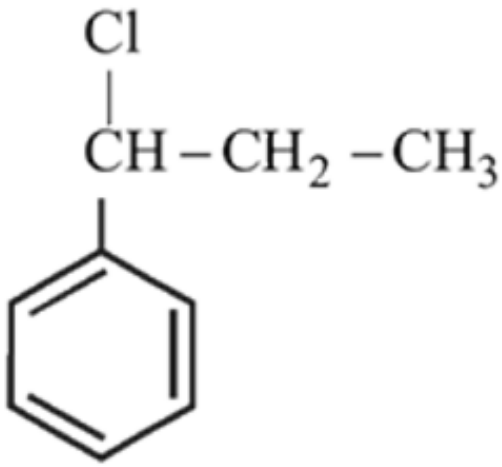
Q.14 Answer:

$-Cl$

Solution:

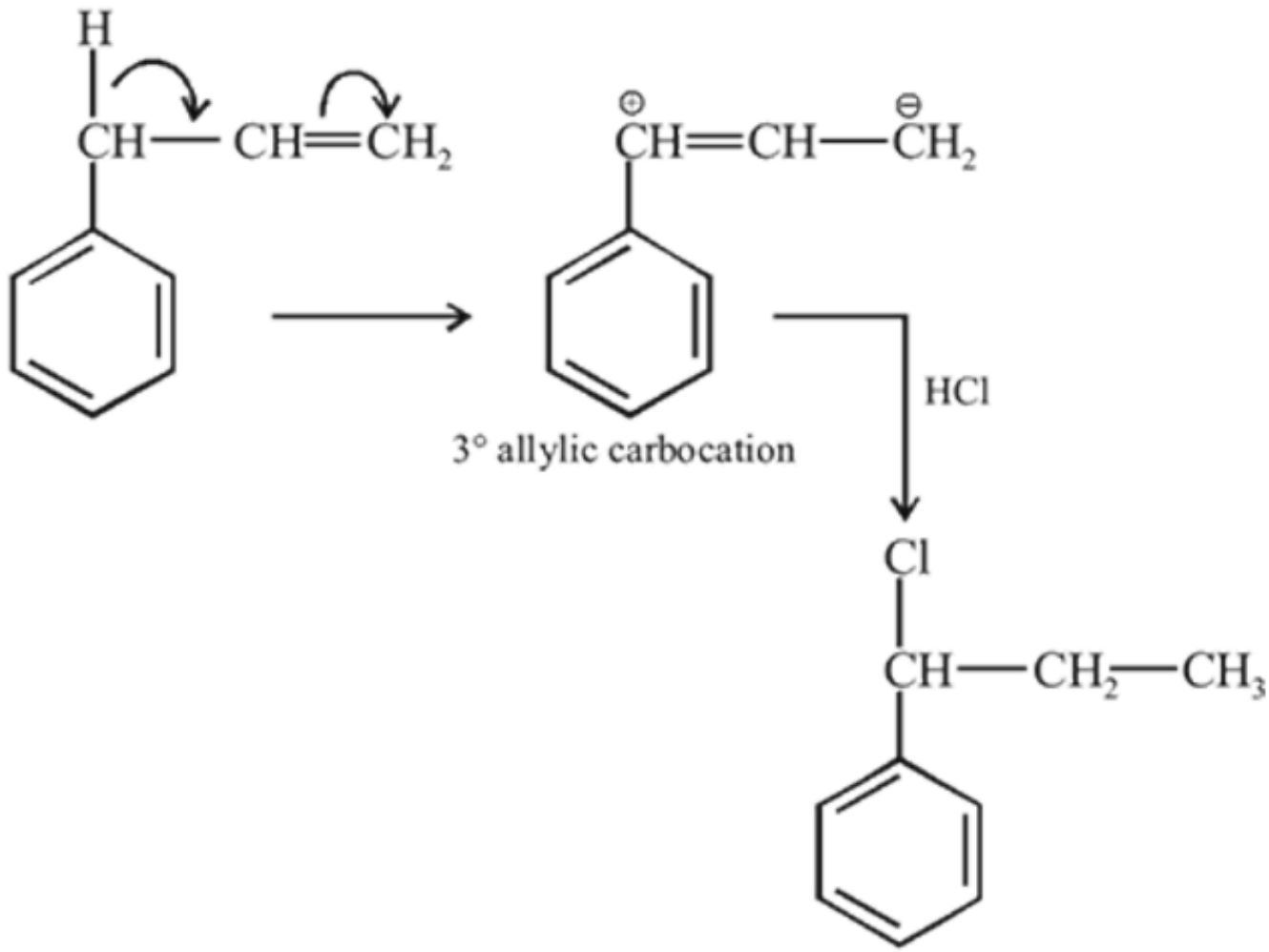
$-Cl$ group is o-, p-directing due to $+R$ effect; however it is deactivating due to strong $-I$ effect of Cl (difference from other o-, p -directing groups which are activating). The net result is that chlorobenzene undergoes o-, p-substitution, but with difficulty.

Q.15 Answer:



Solution:

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(Since tertiary carbocation is more stable)

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